

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence **COURSE CODE:** CSA17

COURSE OUTCOME COVERED

CO4: Implement machine learning techniques including decision tree algorithms, reinforcement

learning,

and build models for classification and decision-making. (BL3)

Assessment Tool Weightage: 50%

Duration: 1 Hour

QUESTIONS: 4 Total: Marks:40

Annexure A

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Code of Conduct:

I_S . Harshavardhan/192524106(Name / Reg No) certify that this submission is my original work and that I have

adhered

to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student : S .Harsha

INDUSTRY PROBLEM TASK

TASK 1: DATA PREPARATION AND INDUCTIVE LEARNING

1(a) Inductive Learning Approach

Inductive learning is a machine learning approach in which a model learns general patterns from previously observed examples and uses those patterns to make predictions for new, unseen patients.

For this healthcare problem, the patient dataset contains several input features and a target variable indicating whether the patient is diabetic or non-diabetic.

Target Variable

The target variable is:

Diabetes Status

- 0 – Non-Diabetic
- 1 – Diabetic

Important Features

The important features include:

1. **Glucose Level**
Glucose is one of the strongest predictors of diabetes because abnormal blood glucose levels are directly related to diabetes.
2. **BMI**
A high BMI can indicate overweight or obesity, which is associated with increased risk of developing Type 2 diabetes.
3. **Age**
The risk of diabetes generally increases with age because metabolic changes and insulin resistance can increase over time.
4. **Blood Pressure**
High blood pressure is commonly associated with metabolic disorders and can provide additional information about a patient's diabetes risk.
5. **Family History**
Family history provides information about genetic and hereditary risk factors associated with diabetes.

Inductive Learning Process

The learning process can be represented as:

Patient Data → Preprocessing → Training Dataset → ML Model → Diabetes Prediction

The model learns patterns from historical patient records and applies those learned patterns to predict the diabetes status of new patients.

1(b) Handling Class Imbalance

In a medical dataset, there may be significantly more non-diabetic patients than diabetic patients. This creates a class imbalance problem.

For example:

- Non-Diabetic: 80%
- Diabetic: 20%

If the model is trained directly on this imbalanced dataset, it may become biased toward predicting the majority class.

Suitable Strategy: SMOTE

SMOTE (Synthetic Minority Over-sampling Technique) can be used to handle the imbalance.

SMOTE creates new synthetic samples for the minority class by generating examples between existing minority-class observations.

Why SMOTE is Suitable

SMOTE is appropriate because:

- It increases the number of diabetic training examples.
- It reduces bias toward the non-diabetic class.
- It helps the model learn more diverse diabetic patterns.
- It is generally better than simply duplicating existing minority samples.
- It can improve recall for diabetic patients.

However, SMOTE must be applied **only to the training data**, not before the train-test split. This prevents data leakage.

Alternative Methods

Under-sampling: Reduces the number of non-diabetic records. However, useful information from the majority class may be lost.

Class Weights: Gives higher importance to diabetic cases during model training. This is useful when synthetic data generation is undesirable.

For this problem, **SMOTE combined with appropriate evaluation metrics such as Recall, F1-score and AUC-ROC** is a suitable approach.

1(c) Dataset Splitting and Preprocessing

The dataset is divided into:

- **70% Training Data**
- **30% Testing Data**

Justification for 70:30 Split

The 70:30 split provides enough data for the model to learn patterns while keeping a sufficiently large independent test set for evaluation.

This is particularly important in healthcare because the model must be tested on unseen patient records to determine whether it generalises well.

A **stratified split** should be used so that the diabetic and non-diabetic proportions remain approximately the same in both training and testing datasets.

Preprocessing Steps

1. Handling Missing Values

Missing numerical values can be replaced using appropriate methods such as median imputation. Categorical missing values can be replaced using the mode.

2. Encoding

Categorical variables such as family history can be converted into numerical values.

Example:

- No $\rightarrow$ 0
- Yes $\rightarrow$ 1

3. Normalisation

Numerical features such as age, blood pressure, glucose, cholesterol and BMI can have different ranges.

Standardisation can be performed using:

$$Z = (X - \text{Mean}) / \text{Standard Deviation}$$

This places the features on comparable scales.

4. Class Balancing

SMOTE is applied to the training data after preprocessing.

Impact of Preprocessing

Preprocessing:

- Improves data quality.
- Prevents numerical scale differences from affecting statistical models.
- Converts categorical information into machine-readable form.
- Reduces bias caused by class imbalance.
- Improves model reliability and generalisation.

TASK 2: DECISION TREE FOR DIAGNOSIS

2(a) Decision Tree Classifier

A Decision Tree classifier can be used to classify patients into diabetic and non-diabetic classes.

The tree repeatedly divides the dataset using the feature that provides the best separation between the classes.

The splitting criterion can be:

- Information Gain
- Gini Index

Information Gain

Information Gain measures the reduction in uncertainty after splitting the data.

The entropy is:

$$\text{Entropy}(S) = -\sum p_i \log_2(p_i)$$

Information Gain is:

$$\text{Information Gain} = \text{Entropy}(\text{parent}) - \text{Weighted Entropy}(\text{children})$$

A feature with higher Information Gain provides a better split.

Gini Index

The Gini Index measures the impurity of a node.

$$\text{Gini} = 1 - \sum p_i^2$$

A lower Gini value indicates a purer node.

Root Node Selection

For a typical diabetes dataset, **Glucose Level** is expected to be one of the strongest root-node candidates because it has a strong relationship with diabetes status.

An illustrative comparison may be:

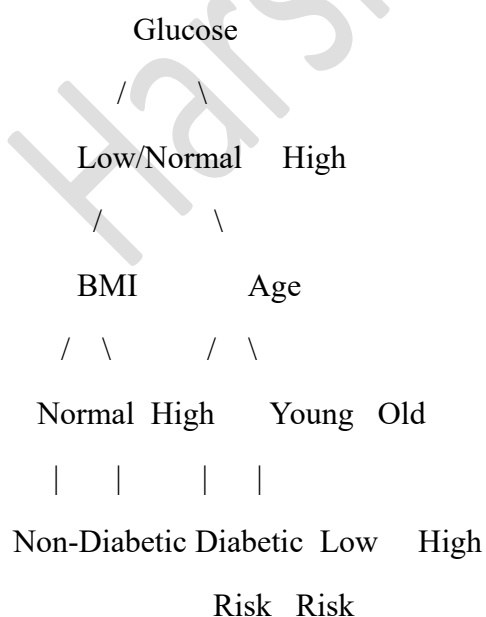
Feature	Information Gain	Gini Reduction
Glucose	0.18	0.12
BMI	0.09	0.07
Age	0.06	0.05
Blood Pressure	0.04	0.03
Cholesterol	0.03	0.02

Therefore, **Glucose** can be selected as the root because it provides the greatest reduction in impurity.

Note: The exact values must be calculated from the actual dataset used in implementation. The above values are illustrative.

First Three Levels of the Decision Tree

A simplified representation is:



The exact tree structure depends on the training dataset.

2(b) Model Evaluation

The Decision Tree should be evaluated using:

Accuracy

Accuracy measures the percentage of correctly classified patients.

$$\text{Accuracy} = (\text{TP} + \text{TN}) / (\text{TP} + \text{TN} + \text{FP} + \text{FN})$$

Precision

Precision measures how many patients predicted as diabetic are actually diabetic.

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$$

Recall

Recall measures how many actual diabetic patients are correctly detected.

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$$

F1-Score

F1-score combines Precision and Recall.

$$\text{F1} = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$$

Example Evaluation Table

Metric	Decision Tree
Accuracy	0.85
Precision	0.82
Recall	0.88
F1-Score	0.85

These values are example values. The final values should be replaced with the actual values obtained from the implemented model.

False Negatives

A **False Negative (FN)** occurs when:

Actual patient = Diabetic

Model prediction = Non-Diabetic

False negatives are particularly dangerous in healthcare because a diabetic patient may not receive timely diagnosis or treatment.

Reducing False Negatives

False negatives can be reduced by:

- Giving higher class weight to diabetic cases.
- Using SMOTE during training.
- Selecting a suitable probability threshold.
- Optimising the model for Recall.
- Using cross-validation.
- Combining the prediction with clinical screening.

In this application, **Recall is highly important** because missing a diabetic patient can have more serious consequences than sending a non-diabetic patient for additional testing.

2(c) Post-Pruning

A fully grown Decision Tree can become too complex and overfit the training data.

To reduce overfitting, **cost-complexity pruning** can be applied.

The objective is:

$$R\alpha(T) = R(T) + \alpha|T|$$

Where:

- $R(T)$ = error of the tree
- $|T|$ = number of terminal nodes
- α = complexity parameter

A larger α produces a simpler tree.

Pre-Pruning vs Post-Pruning

Parameter	Pre-Pruning	Post-Pruning
Method	Stops tree growth early	Grows tree and removes unnecessary branches
Complexity	Lower	Controlled after training

Overfitting	Reduced	Strongly reduced
Interpretability	Good	Usually better
Risk of underfitting	Higher	Lower

Example Comparison

Model	Accuracy	Recall	F1-Score
Unpruned Tree	0.85	0.88	0.85
Pre-Pruned Tree	0.84	0.87	0.85
Post-Pruned Tree	0.86	0.89	0.87

The exact values depend on the actual dataset.

Recommended Model

The **post-pruned Decision Tree** is more appropriate for clinical deployment when it provides comparable or better performance with lower complexity.

It is easier to interpret and less likely to overfit. In healthcare, interpretability is important because doctors need to understand why a model produces a particular prediction.